Teacher/Technician Notes

- Use fresh fruit (strawberry or kiwi works well). You may want this to peeled and chopped into chunks before the lesson to avoid the need for peelers and knives in the lab.
- Check for any fruit allergies to ensure you order fruit which is safe for all of your class to use.
- Pineapple juice is used as it contains an enzyme which removes the proteins from the DNA. If this is unavailable, contact lens cleaning solution may be used as an alternative.
- You may choose to mush the fruit in polystyrene cups to prevent damage to the glassware (or just if you don't like the noise created by metal forks scraping glass!).
- First, test the fruit you are using with the method. If you find that the pulp is too thick to be filtered in a filter funnel, then you may wish to use a sieve lined with kitchen roll and press the liquid out using the back of the fork instead.
- Ethanol (alcohol) should be stored in the freezer and then kept on ice in the lab – ideally this should be requested halfway into the practical to ensure it remains ice cold.

Extracting DNA from Fruit

Aim: To extract DNA from cells.

Equipment:

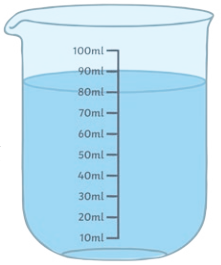
- fruit - peeled and chopped into chunks
- pineapple juice
- table salt
- washing-up liquid
- ice-cold alcohol
- 2 beakers
- boiling tube
- boiling tube rack
- measuring cylinder
- funnel
- filter paper
- glass rod
- fork
- spatula
- pipette

Safety Information

Glassware – risk of cuts if broken.

Apply pressure to wounds and inform teacher immediately.

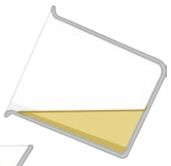
Method:



Step 1: Roughly measure 80ml water into a beaker. Add 1 spatula of table salt and stir well with the glass rod, until all the salt has dissolved.

Step 2: Add 10ml of washing-up liquid to the same beaker and stir gently to mix together. Put this beaker to one side.

Step 3: Into the second beaker, place the chopped fruit and mush it carefully with a fork.



Step 4: Pour the contents of the first beaker (from steps 1 and 2) into the second beaker with the fruit pulp.

Step 5: Stir it a little to mix the contents then leave at room temperature for 20 minutes.

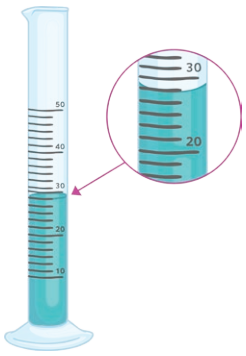


Step 6: Rinse and clean the empty beaker.

Step 7: Set up the filter funnel over the clean beaker.

Step 8: Filter the fruit pulp mix. The filtrate should be a slightly coloured liquid. (The colour of the liquid will depend on the type of fruit you used).

Step 9: Discard the filter paper and pour the filtrate into a measuring cylinder.



Step 10: Add pineapple juice to the filtrate in a ratio of 1:5 pineapple juice to filtrate.

Step 11: Leave at room temperature for 5 minutes.

Step 12: Pour out the mixture from the measuring cylinder into a boiling tube and stand in the boiling tube rack.



Step 13: Carefully pour or pipette ice-cold alcohol down the inside of the boiling tube, in a ratio of 1:1.

Step 14: Leave to separate for 5 to 10 minutes before observing the DNA.